

Pranshu Bansal

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Academic Overview

- **B.S Electrical Engineering & Computer Science (EECS)**
University of California, Berkeley (Class of 2022), GPA: 3.87/4
 - **Coursework:** Deep Learning, Machine Learning, Artificial Intelligence, Financial Engineering, Time Series Analysis, Algorithms, Discrete Math, Probability & Stochastic Processes, Convex Optimization, Econometrics, Data Structures, Computer Architecture, Linear Algebra, Poker Theory
 - **Senior Teaching Assistant**, Discrete Math and Probability Theory for 800+ Students ([Course syllabus](#)) ([Content](#))
- **Intermediate level, Chartered Accountancy, Institute of Chartered Accountants of India** ([National rank 50](#))
- **Bachelor of Commerce (Accounting & Finance)**, Indira Gandhi National Open University, Class of 2018

Programming Skills

- **Languages** - Python, Java, C/C++, MATLAB, Octave, R, SQL
- **ML/Data Science skills** - PyTorch, Google AutoML, Scikit Learn, Numpy, Pandas, Matplotlib

Work Experience

- **Self employed - Options trader, Land trader, Poker player** Feb 2022 - Present
 - Trading equity options, trading in Haryana agricultural land, and playing poker online full time for the last 8 months.
- **Quantitative Trading Analyst - DRW, Chicago** July 2022 - Jan 2023
 - Reengineered the codebase for the market simulation and scenario analysis algorithm reducing runtime by 8x by parallelization. Built a bookkeeper for modularizing and checkpointing the market simulation algorithm for power markets to reduce failure rates by 70%.
 - Analyzed the correlations between market flow due to front month roll of commodities funds and calendar spreads.
 - Built data scraping, aggregating & databasing pipelines to facilitate quick extraction and storage of 200+ GB of market data from 20+ sources.
- **Quantitative Trading Intern - IMC Trading, Chicago** June - Aug 2021
 - **Project 1:** Conducted cross correlations analysis in the commodities markets, identifying strong lead indicators predicting lagged behavior with **89% accuracy**. Computed average latency and designed an algorithm to take favorable positions using indicator signals.
 - **Project 2:** Built a random forests classifier to compute optimal hedge time for options positions in commodities markets using realized volatility signals, momentum indicators, and market liquidity metrics, improving instant profits on a **7-month backtest by 550%**.
 - **Other skills obtained:** Attended 10+ lectures on options theory and market structure, played several mock trading and algorithmic market making games, won second place in fantasy baseball (game theory activity), played 10 hours of poker per week.
- **Quantitative Trading Intern - Sherpa Trading, Austin** Jan - May 2021
 - Developed and parametrized equity trading strategies to piggyback on overnight pump and dump of penny stocks, with a 231% return on 5-year backtests, daily win rate of 87%, and **net edge of 23% per share**.
- **Cofounder & CTO, DeWaste, Inc., Berkeley** ([Website](#)) Nov 2018 - April 2022
 - Generating insights to reduce waste at restaurants using **computer vision, weighing scale, sensors and Raspberry Pi**.
 - Piloted at YUM Brands, UC Berkeley & UC Riverside. Part of YCombinator Startup School, Amp Accelerator, Skydeck; raised **\$25,000 in equity free funding**. Recruited and currently managing 14 interns (ML, data science, web dev, app dev, mech. eng., business dev.)
- **Two years of taxation consulting and financial analyst experience** Aug 2016 - Dec 2018

Selected Projects & Publications

- **Pricing model for valuation of agricultural land in rural areas near Delhi, India**
 - Developed a multifactor model to assist land traders in identifying severe mispricing and trading opportunities in an inefficient land market.
 - **TrashCam: Automatic waste sorting system** ([DevPost](#))
 - Used **Tensorflow CV** model to classify waste (Recycle, Compost or Landfill) in a hardware prototype of a smart trash can using **Raspberry Pi**, cameras and servos. Won **5 prizes** at LAHacks 2019. Made a **ReactJS** web app and a **React Native** mobile app displaying trash analytics.
 - **MIME: A social network for memes** ([Play Store](#)) ([Project Report](#))
 - Meme browser that creates user profiles on the basis of likes. Matches users who have the maximum overlap in interests.
 - Worked on app development, interface design and user research. Drafted project abstract in **LaTeX**.
 - **BeatDrop: Front End Prototype of a music based social media platform** ([Medium link](#))
 - Second prize at Design 360, Esya 2017 of IIIT, Delhi. Models a social media platform based on musical preferences of users.
 - **Publications and professional contributions**
 - Published articles about tax reforms in India on popular online portals like [The Quint](#), [TaxGuru](#) and [CAClubIndia](#).
 - Contributed 5 chapters in the CA textbook "Goods & Services Tax" by CA Vijender Aggarwal. ([Amazon link](#)).
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